MATH 415 - FALL 2020 - EXTRA PRACTICE PROBLEMS - WEEK 10

1. *Which of the following matrices are diagonalizable?*

(1) $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

(2) $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}$

(3) $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

(4) $\begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 2 \\ 0 & 0 & 2 & 3 \end{bmatrix}$

(5) $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$

2. *True/False. Give reasons why! Below A and B are matrices of an appropriate size.*

(1) If A is row equivalent to B, then they have the same eigenvalues.

(2) If A is diagonalizable, then A^2 is diagonalizable.

(3) If all the eigenvalues of the matrix A are zero, then A is the zero matrix.

(4) If A is an $n \times n$ matrix with n different eigenvalues, then A is diagonalizable.

(5) Every triangular matrix has an eigenbasis.

3. *Give a formula for A^n for the following matrix A (Hint: use diagonalization):*

$$A = \begin{bmatrix} 4 & 1 \\ 3 & 2 \end{bmatrix}$$

4. (a) *Let A and B be $n \times n$ matrices. If B is an invertible matrix, does $B^{-1}AB$ always have the same eigenvalues as A?*

(b) (Challenge problem) If A is an $n \times n$ matrix, and the eigenspace for $\lambda_1 \in \mathbb{R}$ has dimension d, show that the characteristic polynomial $p(\lambda) = \det(A - \lambda I)$ of A contains $(\lambda - \lambda_1)^d$ as a factor. (This is why when λ_1 with an eigenvalue of A with multiplicity m, its eigenspace cannot have dimension bigger than m.)

5. *Consider the vectors* $\mathbf{u}$ *and* $\mathbf{v}$ *in* $\mathbb{R}^3$*:*

$$\mathbf{u} = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$$

Decompose $\mathbf{v}$ *into a sum of two vectors, one which points in the same direction as* $\mathbf{u}$ *and one that is orthogonal to* $\mathbf{u}$*.*

6. *Let* $\mathbf{v}$ *be a nonzero vector in* $\mathbb{R}^n$*. A* ***reflector*** *is a matrix* R *such that* $R\mathbf{x} = -\mathbf{x}$ *if* x *is a scalar multiple of* $\mathbf{v}$*, and* $R\mathbf{x} = \mathbf{x}$ *if* $\mathbf{v} \cdot \mathbf{x} = 0$*.*

(1) Someone tells you that '*When applying a reflector, the orthogonal complement of* span($\mathbf{v}$) acts as a mirror.' Can you explain what that person meant? Draw a picture!

(2) Show that $R = 2P - I$, where P is projection matrix for projection onto the orthogonal complement of span($\mathbf{v}$). (If you haven't learned about projections onto arbitrary subspaces, then just consider the case $n = 2$).

(3) Deduce from *(2)* that $R = I - \frac{2}{\mathbf{v} \cdot \mathbf{v}} \mathbf{v}\mathbf{v}^T$.

(4) Deduce from *(3)* that $R^T = R$ and $R^T R = I$.